

First addendum to SOLPS-ITER 3.0.9/3.1.1 Release Notes

May 3rd, 2024

Following various user suggestions and the return on experience from the SOLPS-ITER workshop held last month at Dalian University of Technology (again, many thanks to [Sang Chaofeng EXT](#) and his team for a very pleasant time in Dalian and a most productive workshop), I would like to inform you of several improvements that have been made to the code distributions and that are now available to you by means of the usual update scripts (*solps-iter_update* for the *master* branch and *solps-iter_update_develop* for the *develop* branch) and/or from the GIT repository directly. These also apply to the *release/3.1.1* code branches.

These improvements are:

- Following the practice established in the EDGE2D-EIRENE code, a new pre-processor pragma -DLEGACYCOMP can be used to indicate if your compiler does not support the Fortran 2003 standard. This replaces the -DF2003 pragma used before that had the opposite meaning. The configuration files have been modified accordingly. Please note that the **Eirene** MsV contributing rules expect new code additions to follow the Fortran 2003 standard, so it may not always be possible to maintain backward compatibility with older compilers, though I will try to provide alternatives whenever possible.
- For ASIPP users that are running on the NSCC cluster, a new set of configuration files for the NSCC domain have been created which you should use instead of locally modifying the ASIPP ones. Simply include the command "setenv HOST_NAME NSCC" inside your *\$SOLPSTOP/SETUP/setup.csh.HOST_NAME.local* file (create it if necessary) to indicate you are running on the NSCC cluster. I have also fixed a number of lingering issues with the ASIPP configuration files that led to annoying warning messages during compilation. In particular, the **B2.5 Makefile** no longer relies on the basename command and the condition for linking to the IMAS modules has been corrected.
- Some conditioning of compiler options has been added when needed if those options are not recognized by older compiler versions.
- In some cases, the serial compiler is Intel-based while the MPI compiler is GNU-based. The **Uinp Makefile** can now deal with this, as well as the **B2.5** and **Eirene** ones.
- Some more DLUT configuration files have been added.
- In **Uinp**, the rule for adding a time census stratum to the *b2.neutrals.parameters.stencil* file has been corrected so it only appears if one is also defined in the **Eirene** input file.
- The *carre* script now includes checks to verify that the preparation and gridding steps did produce the expected output files before allowing the user to proceed.
- A new main *Makefile* target alias, *solps_nox*, can be used to ask to build the standard set of SOLPS executables without graphics. The clean and debug versions, *clean_solps_nox* and *solps_nox_debug*, are also available.
- Please note that the CMake version required by the **Eirene** MsV has been upped to 3.0 from 2.8. I also included means to include a local addition to the CMake instructions files using a *\$SOLPSTOP/modules/Eirene/src.local/CMakeLists.txt* file.

The centralized installations of the 3.0.9 *master* and *develop* code versions on the ITER SDCC cluster (which you will find at */work/projects/solps-iter/3.0.9_master* and */work/projects/solps-iter/3.0.9_develop* respectively) as well as on the JFRS-1 machine at IFERC (at */project/SOLPS/solps-iter* and */project/SOLPS/solps-iter_develop*) have been updated and recompiled to reflect the changes mentioned above.